

Coordination Compounds

Complete notes — text, 24 redrawn diagrams and typeset tables

About this document. This is the full text of the source notes, in its own order, with every diagram **redrawn** as sharp vector artwork and every table **typeset**, in place of the low-resolution screenshots. Redrawing meant checking, and the source carries **nine mistakes** — each is corrected here and flagged in red where it occurs:

1. Fig 5 — “square bipyramidal” should be **square pyramidal**.
2. Fig 16 — the optical pair must be the **cis** isomer; the trans form is achiral.
3. Fig 17 — square planar is **dsp²**, not dsp³.
4. Fig 22 — the hybrid is **sp³d²**, not sp²d².
5. Fig 7 — nitrate donates through **O**, not N.
6. Fig 8 — carbon monoxide donates through **C**, not O.
7. Fig 9 — sulphate donates through **O**, not S.
8. Spellings: *hydrido*, *fluorido*, *phenanthroline*.

Important Topics

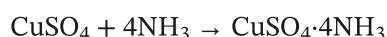
The concept of coordination compounds originates from the tendency for the complex formation of the transition elements.

Complex compounds or coordination compounds are those **molecular compounds that retain their identity in solid and in solution** are known as complex compounds.



Molecular or Addition Compounds:

When solutions of two or more simple stable compounds in molecular proportion are allowed to evaporate, crystals of new substances, called molecular or addition compounds, are obtained. For Example:



There are two types of molecular or addition compounds: Double salts or lattice compounds and Coordination or complex compounds.

(1) Double salts or lattice compounds: These are the molecular compounds that exist only in the solid-state and lose their identity when dissolved in water, i.e., they dissociate into simple ions completely when dissolved in water. For Example $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ (Mohr's Salt), $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (Potash Alum).

(2) Coordination or Complex Compounds: These are the molecular compounds that retain their identity in the solid as well as in the dissolved state and their properties are completely different from those of their constituent particles. A part of these compounds is not dissociated in solution and its behavior is different from its constituents. For example, potassium argentocyanide does not form Ag^+ and CN^- simpler ions but instead, gives argentocyanide complex ion $[\text{Ag}(\text{CN})_2]^-$ (which does not dissociate).

Cationic complexes: A complex in which the complexion carries a net positive charge is called a cationic complex. For Example, $[\text{CoCl}_2(\text{en})_2]^+$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$.

Anionic complexes: A complex in which the complexion carries a net negative charge is called anionic complex, e.g., $[\text{Ag}(\text{CN})_2]^-$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$.

Neutral complexes: A complex carrying no net charge is called a neutral complex or simply a complex e.g., $[\text{Ni}(\text{CO})_4]$, $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{Ni}(\text{CO})_4]$.

Some Important Definition

1) Coordination Entity: Central atom/ion (metal) to which a fixed number of other atoms or groups are attached (ligand). For example, $[\text{CoCl}_3(\text{NH}_3)_3]$ is a coordination entity in which the cobalt ion is surrounded by three ammonia molecules and three chloride ions. Other examples are $[\text{Ni}(\text{CO})_4]$, $[\text{PtCl}_2(\text{NH}_3)_2]$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$.

A coordination entity can be neutral or charged.

2) Central atom/ion: In a coordination entity, the atom/ion to which a fixed number of ions/neutral molecules are attached is called the central atom or ion. For example, the central atom/ion in the co-ordination entities: $[\text{NiCl}_2(\text{H}_2\text{O})_4]$, $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ are Ni^{2+} , Co^{3+} and Fe^{3+} respectively.

These central atoms/ions are also referred to as Lewis acids since they accept electron pairs from ligands.

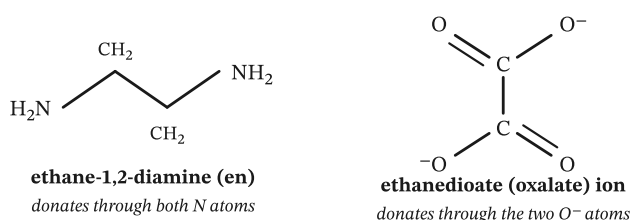


Fig 1 Bidentate ligands

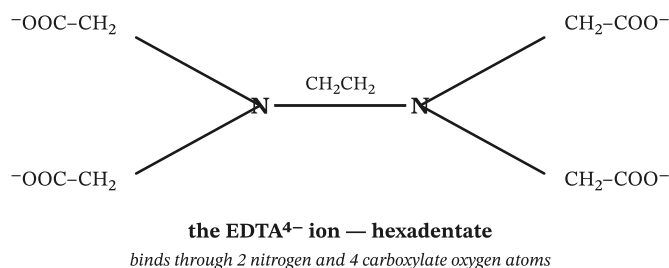


Fig 2 The EDTA⁴⁻ ion

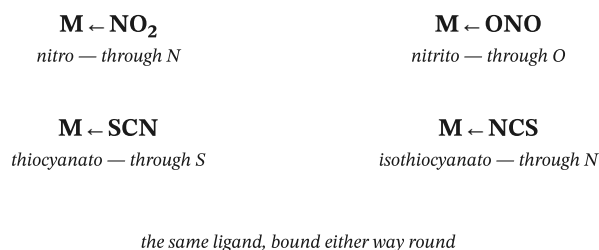
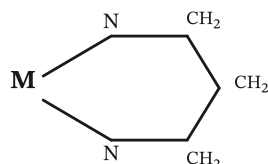


Fig 3 Ambidentate ligands



a five- or six-membered chelate ring
every polydentate ligand is a chelating ligand

Fig 4 A chelate ring

4) Co-ordination number: The number of atoms in a ligand that directly bond to the central metal atom or ion by coordinate bonds is called the coordination number of the metal atom or ion. Some common coordination numbers exhibited by metal ions are 2, 4, and 6. The light transition metals exhibit 4 and 6 coordination numbers while heavy transition metals exhibit coordination numbers above 6.

For example, in the complex ion $[\text{PtCl}_6]^{2-}$ the coordination number of Pt is 6 and in $[\text{Ni}(\text{NH}_3)_4]^{2+}$, the coordination number of Ni is 4. Similarly, in the complexions, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{Co}(\text{en})_3]^{3+}$, the coordination number of both Fe and Co, is 6 because $\text{C}_2\text{O}_4^{2-}$ and en (ethane-1,2- diamine) are bidentate ligands.

5) Co-ordination sphere: The central metal atom/ion and the ligands directly attached to it are collectively termed as the coordination sphere. The coordination sphere is represented inside square brackets, e.g. $[\text{Fe}(\text{CN})_6]^{4-}$.

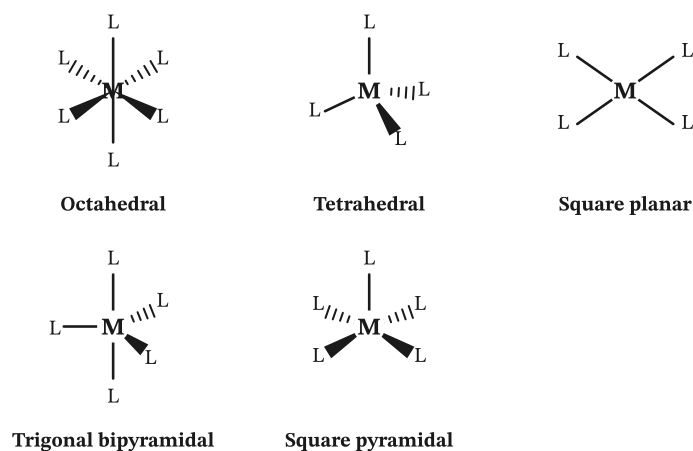


Fig 5 Coordination polyhedra

Correction · The source figure labels the five-coordinate shape “square **bipyramidal**”. It is **square pyramidal** — five ligands, a square base and one apex. A bipyramid on a square base has six, which is the octahedron.



Let the oxidation number of cobalt be x . Water and ethane-1,2-diamine are neutral, the cyanide is -1 , and the whole ion carries $+2$:

$$x + 0 + (-1) + 2(0) = +2$$

$$x - 1 = +2 \Rightarrow x = +3$$

The coordination number is 6: one O from water, one C from cyanide and two N from each of the two en.

Fig 6 Finding the oxidation number

8) Homoleptic and Heteroleptic Complexes: Complexes in which the central metal atom or ion is linked to the only types of ligands are called **homoleptic complexes**. For example, $[\text{Co}(\text{NH}_3)_6]^{3+}$.

The complexes in which the central metal atom or ion is linked to more than one kind of ligand are called **heteroleptic complexes**. For example, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

Nomenclature of Coordination Compounds Class 12

Werner's Theory of Coordination Compounds

Alfred Werner, a Swiss chemist proposed a theory about the nature of bonding in the coordination compounds (complexes) in the year 1893. The main postulates of this theory are:

1. Metals possess two types of linkages (valencies).
 - a) Primary or principal or ionizable links (valency) which are the same as the oxidation state of the metal.
 - b) Secondary or non-ionizable links (valency) which are the same as the coordination number of the central metal atom/ion. This number is fixed for a metal.
2. A metal atom satisfies both primary as well as secondary linkages (valencies). Primary linkages (valencies) are satisfied by negative ions whereas secondary linkages (valencies) are satisfied by ligands (neutral or negative ions/groups).
3. Ligands satisfying secondary linkages (valencies) are directed towards fixed directions (spatial arrangements) giving a definite geometry to the complex but primary valencies are non-directional. Such spatial arrangements are called coordination polyhedra.

Name	Formula	Charge	Donor atom	Name in the complex
Cyanide ion	CN [−]	−1	C	cyanido (cyano)
Halide ion	X [−] (F, Cl, Br, I)	−1	X	fluorido / chlorido / bromido / iodido
Hydride ion	H [−]	−1	H	hydrido
Nitrite (through N)	NO ₂ [−]	−1	N	nitrito-N (nitro)
Nitrite (through O)	ONO [−]	−1	O	nitrito-O
Nitrate ion	NO ₃ [−]	−1	O	nitrate
Hydroxide ion	OH [−]	−1	O	hydroxido
Amide ion	NH ₂ [−]	−1	N	amido
Thiocyanate ion	SCN [−]	−1	S	thiocyanato-S
Isothiocyanate ion	NCS [−]	−1	N	thiocyanato-N (isothiocyanato)
Acetate ion	CH ₃ COO [−]	−1	O	acetato

Corrected from the source ·

- Nitrate, NO₃[−], donates through **O**, not N — the source table says N.
- Spellings corrected: *hydrido* (printed “hidrido”) and *fluorido* (printed “flourido”).

Fig 7 Anionic ligands

Name	Formula	Charge	Donor atom	Name in the complex
Methylamine	CH ₃ NH ₂	0	N	methylamine
Ammonia	NH ₃	0	N	ammine
Water	H ₂ O	0	O	aqua
Nitric oxide	NO	0	N	nitrosyl
Carbon monoxide	CO	0	C	carbonyl
Thiocarbonyl	CS	0	S	thiocarbonyl
Phosphine	PH ₃	0	P	phosphine
Triphenylphosphine	(C ₆ H ₅) ₃ P	0	P	triphenylphosphine
Thiourea (tu)	(NH ₂) ₂ CS	0	S	thiourea
Pyridine (py)	C ₅ H ₅ N	0	N	pyridine
Ethylenediamine (en)	H ₂ NCH ₂ CH ₂ NH ₂	0	two N	ethane-1,2-diamine
2,2'-Dipyridyl (dipy)	C ₁₀ H ₈ N ₂	0	two N	dipyridyl
o-Phenanthroline	C ₁₂ H ₈ N ₂	0	two N	phenanthroline

Corrected from the source ·

- Carbon monoxide donates through **C**, not O — which is why the ligand is called *carbonyl* and the bond formed is M–C. The source table says O.
- Spelling corrected: *phenanthroline* (printed “phenonanthroline”).

Fig 8 Neutral and bidentate ligands

Name	Formula	Charge	Donor atom	Name in the complex
Oxide ion	O^{2-}	-2	O	oxido (oxo)
Peroxide ion	O_2^{2-}	-2	O	peroxido
Sulphide ion	S^{2-}	-2	S	sulphido
Sulphite ion	SO_3^{2-}	-2	S	sulphito
Sulphate ion	SO_4^{2-}	-2	O	sulphato
Thiosulphate ion	$S_2O_3^{2-}$	-2	S	thiosulphato
Carbonate ion	CO_3^{2-}	-2	O	carbonato
Imide ion	NH_2^-	-2	N	imido
Oxalate ion (ox)	$C_2O_4^{2-}$	-2	two O	oxalato
Acetylacetonate (acac)	$CH_3COCHCOCH_3^-$	-1	two O	acetylacetonato
Glycinate ion (gly)	$NH_2CH_2COO^-$	-1	one N, one O	glycinato
Dimethylglyoximate (dmg)	$C_4H_7N_2O_2^-$	-1	one N, one O	dimethylglyoximato

Corrected from the source ·

- Sulphate, SO_4^{2-} , donates through **O**, not S — the source table says S.

Fig 9 Further anionic and bidentate ligands

Rules for naming mononuclear coordination compounds:

1. Order of naming ions: The positive ion (cation) whether simple or complex, is named first followed by the negative ion (anion). The name is started with a small letter and the complex part is written as one word.

2. Naming ligands:

i) Negative ligands (organic or inorganic) end in -O, eg, CN^- (cyano), Cl^- (chlorido), Br^- (bromido), F^- (fluorido), NO_2^- (nitro), OH^- (hydroxo), O^{2-} (oxo), H^- (hydrido). If the name of the anionic ligands ends in -ide, -ite or -ate, the last 'e' is replaced by 'O' giving -ido, -ito and -ato eg, SO_4^{2-} (sulphato), $C_2O_4^{2-}$ (oxalato), NH_2^- (amido), NH_2^- (imido), ONO^- (nitrito).

ii) Neutral ligands have no special ending, NH_3 (ammine), H_2O (aqua), CO (carbonyl)

iii) Positive ligands (which are very few) end in -ium, NO^+ (nitrosoium), NO_2^+ (nitronium).

3. Numerical prefixes to indicate the number of ligands: If there are several ligands of the same type, the prefixes like di, tri, tetra, penta, and hexa are used to indicate the number of ligands of that type. When the name of polydentate ligand includes a number e.g., ethylenediamine, then bis, tris, tetrakis are used as prefixes.

4. Order of naming of ligands: All ligands whether negative, neutral or positive are named first in alphabetical order followed by the name of the metal atom/ion.

5. Naming of the complexion and ending of the central atom: Ligands are named first followed by the metal atom.

a) If the complexion is a cation or the coordination compound is non-ionic, the name of the central metal ion is written as such followed by its oxidation state indicated by a Roman numeral (such as II, III, IV) in the parentheses at the end of the name of the metal without any space between the two.

b) If the complexion is an anion, the name of the central metal atom is made to end in -ate followed by the oxidation number in brackets without any space between them.

Rules for Writing Formula from the Name of the Mononuclear Complex

(a) Formula of the cation (whether simple or complex) is written first, followed by the anion.

(b) The formula of the complexion (coordination entity) (whether charged or not) is written in square brackets called coordination sphere.

(c) Within the coordination sphere, the symbol of the metal atom is written first followed by the symbols/formulas of the ligands arranged alphabetically according to their names irrespective of the charge present on them. While listing the ligands alphabetically, the following rules should be followed:

i) Polydentate ligands are also listed in alphabetical order.

ii) The position of abbreviated ligands in the alphabetical order is determined from the first letter of the abbreviation.

iii) The position of ligands with special names (such as aqua for water) in alphabetical order is determined from the first letter of the special name.

iv) Abbreviations used for the ligands and the formulas of the polyatomic ligands are enclosed in parentheses separately.

v) The metal atom as well as all the ligands are listed without any space between them.

(d) If the formula of the complexion is to be written without writing the counter ion, the charge on the complexion is indicated outside the square bracket as a right superscript with the number before the sign. For example, $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$, etc.

(e) The number of cations or anions to be written in the formula is calculated on the basis that the total positive charge must be equal to the total negative charge, as the complex as a whole is electrically neutral.

Example: Tetraammineaquachloridocobalt(III) chloride has complex ion $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]^-$ and simple ion = Cl⁻

To balance the charge, the formula will be $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$.

Examples of Nomenclature of Coordination Compounds

1. $[\text{Cr}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{Cl}_3$ – triamminetriaquachromium(III) chloride

Explanation: The complexion is inside the parentheses, which is a cation.

The ammine ligands are named before the aqua ligands according to alphabetical order.

Since there are three chlorides binding with the complexion, the charge on the complexion must be +3 (since the compound is electrically neutral).

From the charge on the complexion and the charge on the ligands, we can calculate the oxidation number of the metal. In this example, all the ligands are neutral molecules. Therefore, the oxidation number of chromium must be the same as the charge of the complexion, +3.

2. $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Br}_3$ – pentaamminechloroplatinum(IV) bromide

Explanation: The complexion is a cation, the counter anion is the 3 bromides.

The charge of the complexion must be +3 since it bonds with 3 bromides.

The NH_3 are neutral molecules while the chloride carries -1 charge. Therefore, the oxidation number of platinum must be +4.

3. $[\text{Pt}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_2\text{Cl}_2]\text{Cl}_2$ – dichlorobis(ethylenediamine)platinum(IV) chloride

Explanation: ethylenediamine is a bidentate ligand, the bis- prefix is used instead of di-

4. $[\text{Co}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_3]_2(\text{SO}_4)_3$ – tris(ethylenediamine)cobalt(III) sulfate

Explanation: The sulfate is the counter anion in this molecule. Since it takes 3 sulfates to bond with two complex cations, the charge on each complex cation must be +3.

Since ethylenediamine is a neutral molecule, the oxidation number of cobalt in the complex ion must be +3.

Again, remember that you never have to indicate the number of cations and anions in the name of an ionic compound.

5. $K_4[Fe(CN)_6]$ – potassium hexacyanoferrate(II)

Explanation: potassium is the cation and the complex ion is the anion.

Since there are 4 K^+ binding with a complex ion, the charge on the complex ion must be -4 .

Since each ligand carries a -1 charge, the oxidation number of Fe must be $+2$.

The common name of this compound is potassium ferrocyanide.

6. $Na_2[NiCl_4]$ – sodium tetrachloronickelate(II)

Explanation: The complex ion is the anion so we have to add the suffix $-ate$ in the name of the metal.

7. $Pt(NH_3)_2Cl_4$ – diamminetetrachloroplatinum(IV)

Explanations: This is a neutral molecule because the charge on Pt^{+4} equals the negative charges on the four chloro ligands.

If the compound is $[Pt(NH_3)_2Cl_2]Cl_2$, even then the number of ions and atoms in the molecule are identical to the example, it should be named: diamminedichloroplatinum(II) chloride, a big difference.

8. $Fe(CO)_5$ – pentacarbonyliron(0)

Explanations: Since it is a neutral complex, it is named in the same way as a complex cation. The common name of this compound, iron carbonyl, is used more often.

9. $[CrCl_2(H_2O)_4]^+$ – tetraaquadichlorochromium(III) ion

Explanation – The ligands here are Cl and H_2O . Therefore, we will use the monodentate ligand names of “chloro” and “aqua”. Alphabetically, aqua comes before chloro, so this will be their order in the complex’s name. There are 4 aqua’s and 2 chloro’s, so we will add the number prefixes before the names. Since both are monodentate ligands, we will say “tetra[aqua]di[chloro]”.

10. $(NH_4)_2[Ni(C_2O_4)_2(H_2O)_2]$ – ammonium diaquabis(oxalato)nickelate(II)

Explanation: The oxalate ion is a bidentate ligand.

11. $[Ag(NH_3)_2][Ag(CN)_2]$ – diamminesilver(I) dicyanoargentate(I).

Some Other Examples:

1. $[NiCl_4]^{2-}$
2. $Pt(NH_3)_2Cl_4$
3. $[Pt(NH_3)_2Cl_2]Cl_2$
4. $[Co(NH_3)_5CO_3]Cl$
5. $[Cr(H_2O)_4Cl_2]Cl$
6. $[Zn(OH)_4]^{-2}$

Answers:

1. Tetrachloridonickelate(II) ion. The complex ion, an anion, is inside the parentheses. We have to add the suffix $-ate$ in the name of the metal.
2. Diamminetetrachloroplatinum(IV). This is a neutral molecule. The total charge on the ligands is -4 . Therefore, the platinum oxidation number is $+4$.
3. Diamminedichloroplatinum(II) chloride. Here, the number of ions and atoms are the same. However, the brackets as well as the different oxidation number of the platinum result in a very different name.
4. Pentaamminecarbonatocobalt(III) chloride.
5. Tetraaquadichlorochromium(III) chloride
6. Tetra hydroxide zincate (II) ion.

Werner's theory could not explain:

- a) Only certain elements possess the remarkable property of forming coordination compounds?
- b) The bonds in coordination compounds have directional characteristics.
- c) Coordination compounds have characteristic magnetic and optical properties.

Isomerism in Coordination Compounds Class 12

Isomerism

Two or more chemical compounds which have identical chemical formula but different structures are known as isomers and the phenomenon is known as isomerism. The isomers have different arrangements of ligands around the central metal atom. The isomerism shown by co-ordination compounds are broadly divided into two:

Structural Isomerism and Stereo Isomerism.

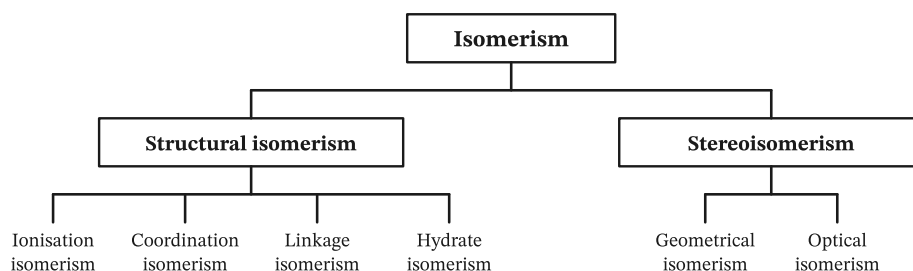


Fig 10 The kinds of isomerism

Structural Isomerism: These are isomers that differ in the structural arrangement of ligands around the central atom. They are of four types:

- (a) **Ionisation Isomerism**
- (b) **Coordinate Isomerism**
- (c) **Linkage Isomerism**
- (d) **Hydrate Isomerism**

(a) Ionisation Isomerism: This form of isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then become the counter ion. Example: $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$,

$[\text{Pt}(\text{OH})_2(\text{NH}_3)_4]\text{SO}_4$ and $[\text{Pt}(\text{SO}_4)(\text{NH}_3)_4](\text{OH})_2$,

$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ and $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$.

(b) Coordinate Isomerism: This type of isomerism arises from the interchange of ligands between cationic and anionic entities of different metal ions present in a complex.

Example: $[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{C}_2\text{O}_4)_3]$ and $[\text{Cr}(\text{NH}_3)_6] [\text{Co}(\text{C}_2\text{O}_4)_3]$,

$[\text{Cu}(\text{NH}_3)_4] [\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4] [\text{CuCl}_4]$

$[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{NH}_3)_6] [\text{Co}(\text{CN})_6]$

(c) Linkage Isomerism: This isomerism takes place when a monodentate ligand has two possible donor atoms and attaches in two different ways to the central metal atom. Such ligands are known as also ambidentate ligands. For example, nitro (NO_2^-) and nitrito ($-\text{ONO}^-$), cyano ($-\text{CN}^-$), and isocyano ($-\text{NC}^-$).

For Example: $[\text{Cr}(\text{H}_2\text{O})_5\text{SCN}]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_5\text{NCS}]^{2+}$, $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ and $[\text{Co}(\text{ONO})(\text{NH}_3)_5]\text{Cl}_2$.

(d) Hydrate Isomerism: It is isomerism in which solvent is involved as ligand. If the solvent is water it is called

hydrate isomerism. This form of isomerism is also known as 'solvate isomerism'.

For Example: $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (violet) and its solvate isomer $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (grey-green).

Stereo-isomerism: In stereo-isomerism, the isomers differ only in the spatial arrangement of atoms or groups about the central metal atom. It is also known as space isomerism. It can be further classified into two types:

(i) Geometrical isomerism (ii) Optical isomerism



Fig 11 Square planar cis / trans

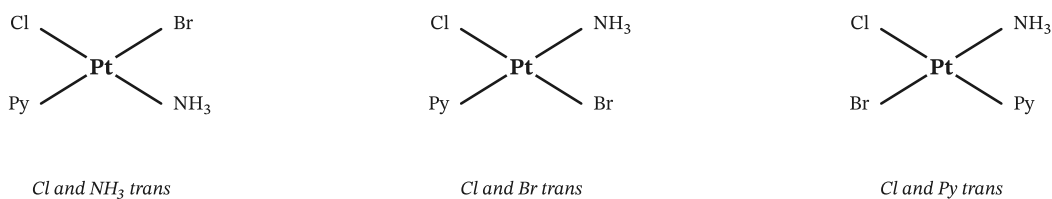


Fig 12 The three isomers of a square planar $[\text{MABCD}]$

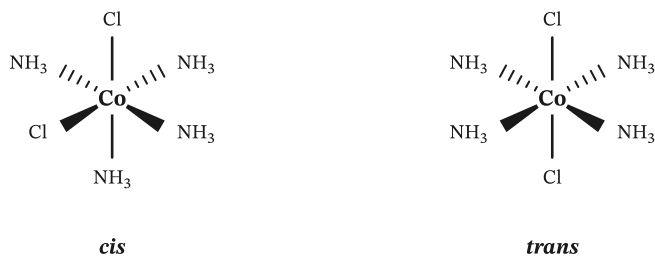


Fig 13 Octahedral $[\text{MX}_2\text{L}_4]$ cis / trans

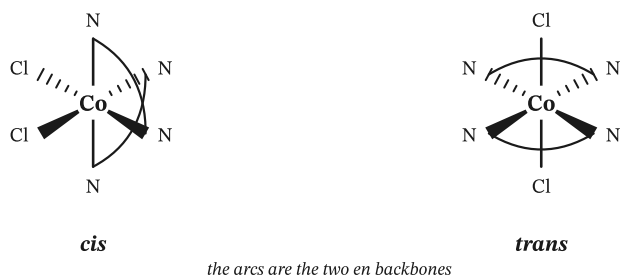


Fig 14 Octahedral $[\text{MX}_2\text{A}_2]$ cis / trans

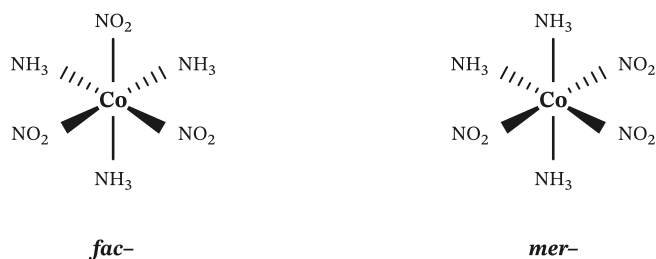


Fig 15 Facial and meridional isomers

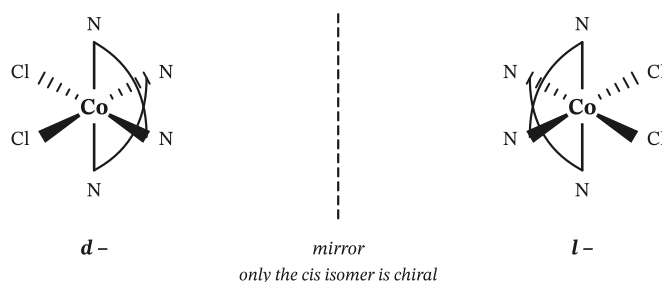


Fig 16 Optical isomers

Correction · The source figure draws the **trans** isomer as the optical pair. Only the **cis** isomer is chiral; the trans form has a plane of symmetry and is optically inactive, so the mirror pair is drawn cis here.

Valence Bond Theory Coordination Compounds

Coordination number	Hybridisation	Geometry
4	sp^3	Tetrahedral
4	dsp^2	Square planar
5	sp^3d	Trigonal bipyramidal
6	sp^3d^2	Octahedral (outer orbital)
6	d^2sp^3	Octahedral (inner orbital)

Corrected from the source ·

- Square planar is **dsp^2** , not dsp^3 — coordination number 4 needs four hybrid orbitals, and $d + s + p + p$ is four. The source table says dsp^3 , which is five orbitals.

Fig 17 Coordination number, hybridisation and geometry

FORMATION OF OCTAHEDRAL COMPLEXES

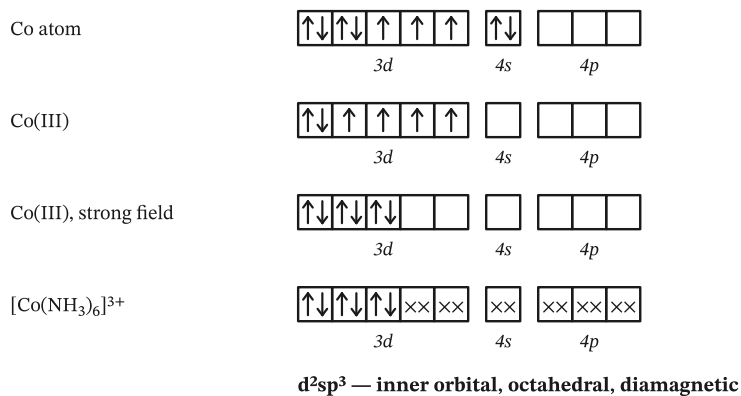


Fig 18 $[\text{Co}(\text{NH}_3)_6]^{3+}$

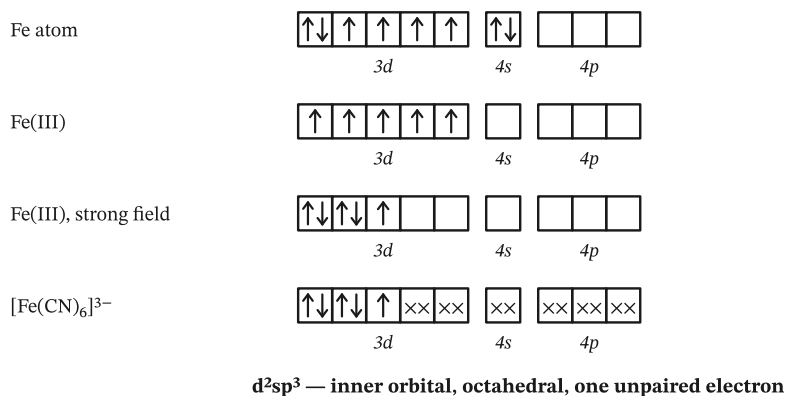


Fig 19 $[\text{Fe}(\text{CN})_6]^{3-}$

FORMATION OF SQUARE PLANAR COMPLEXES

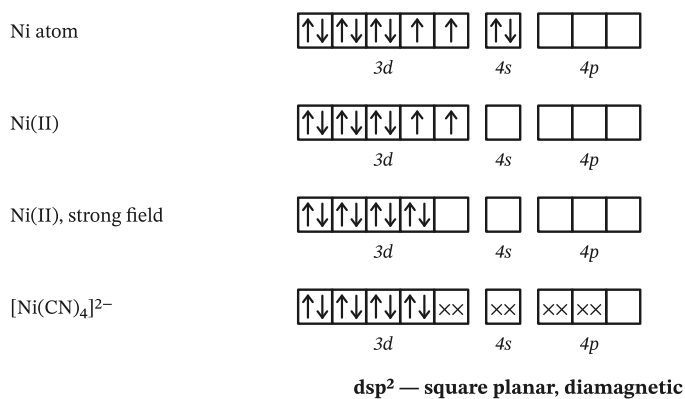


Fig 20 $[\text{Ni}(\text{CN})_4]^{2-}$

FORMATION OF TETRAHEDRAL COMPLEXES

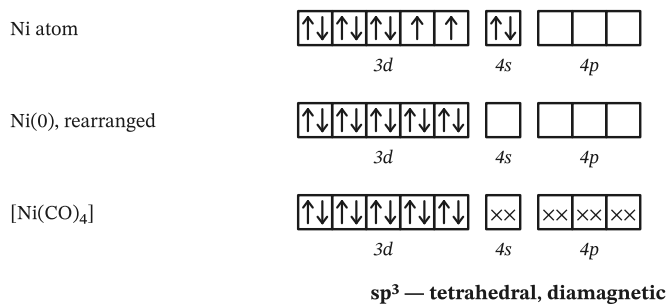


Fig 21 [Ni(CO)₄]

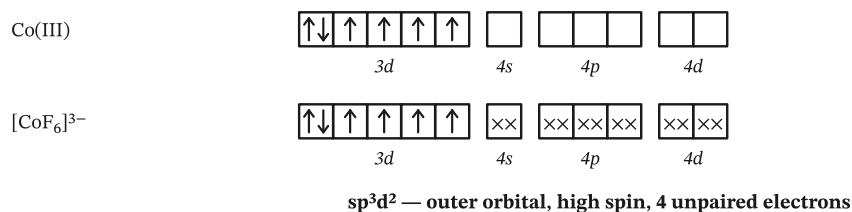


Fig 22 [CoF₆]³⁻

Correction · The source figure labels the hybrid **sp²d²**, which is only four orbitals. Six ligands need six: **sp³d²**.

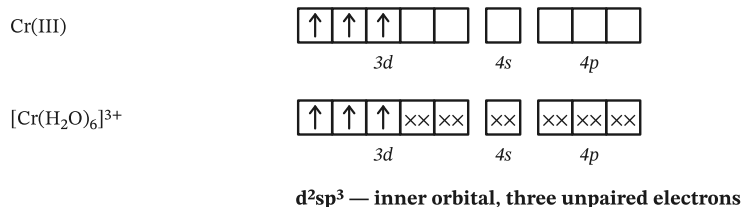


Fig 23 [Cr(H₂O)₆]³⁺

Coordination number	Hybridisation	Shape	Arrangement
2	sp	Linear	X—A—X
4	sp ³	Tetrahedral	four corners of a tetrahedron
4	dsp ²	Square planar	four corners of a square
5	sp ³ d	Trigonal bipyramidal	three equatorial, two axial
6	sp ³ d ² or d ² sp ³	Octahedral	six corners of an octahedron

Fig 24 Common types of hybridisation

Limitations of Valence Bond Theory:

Even though the valence bond theory explains the formation, structures, and magnetic behavior of coordination compounds to a larger extent, it suffers from the following shortcomings:

- i). It does not explain the relative energies of different structures.
- ii). It offers no explanation of the colour observed for the complex.
- iii). It cannot predict whether a 4-coordinate complex will be tetrahedral or square planar.
- iv). It does not take into account the splitting of d-energy levels.
- v. It does not explain the spectra of complexes.

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